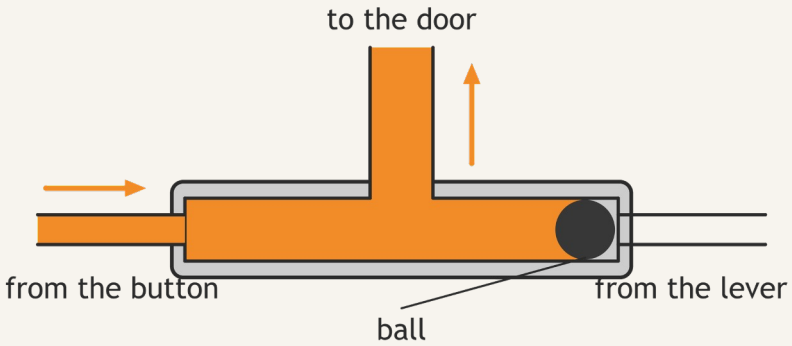
Daily Review

1	Brake off, lever pushed. The door?	Shut
2	Move a roller valve only with…	The air off
3	P = 0.2 N mm^{-2}, A = 300 mm^2. The force?	60 N
4	Port 3 on a 3/2 valve is the…	Exhaust
5	Two switches in series: the lamp is on when…	Both are closed

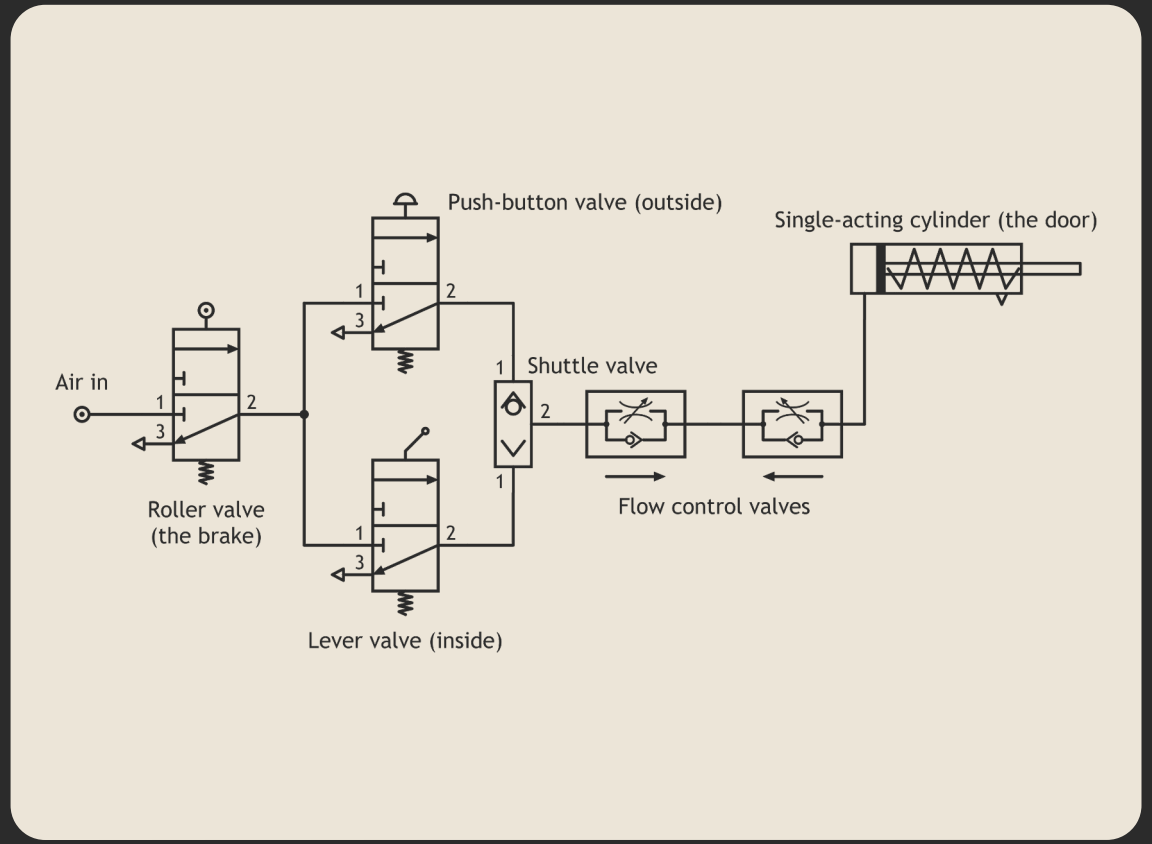
How Does the Door Decide?

Inside the shuttle valve: OR, AND and the brake.



S3 Engineering Science

Brake off, lever pushed: the door stayed shut. Why?



Lesson 7: what you found

By the end of this lesson…



1. I can explain how the brake stops the door
2. I can tell OR control from AND control
3. I can name the roller, roller trip and tee symbols
4. I can describe how the door works, stage by stage

The Roller Valve

Roller

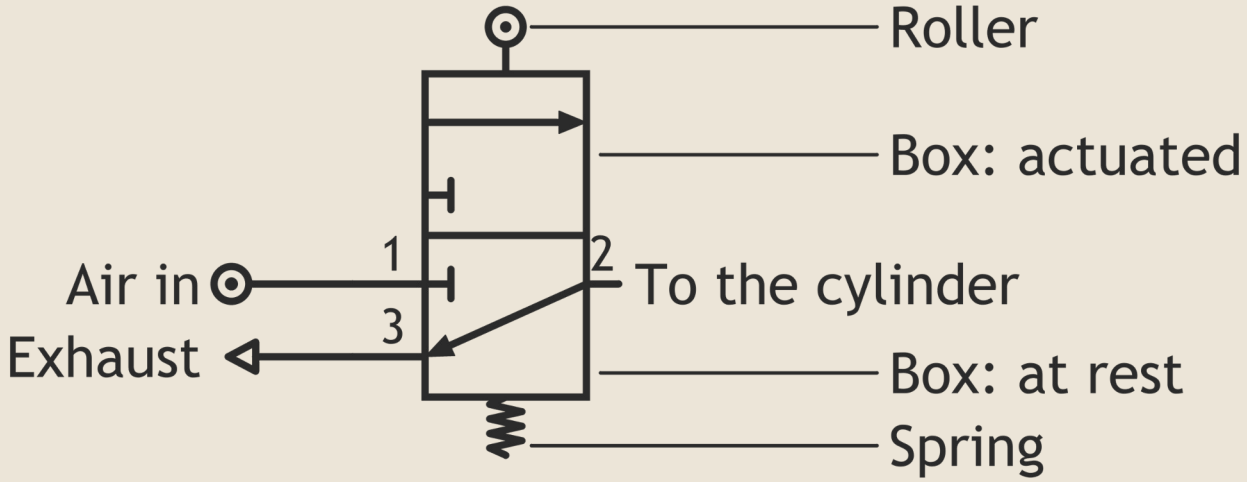
a wheel on a sprung arm

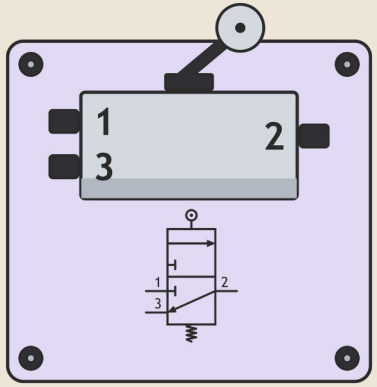
Pressed by

a moving part, e.g. a rod

Symbol

a circle on top





The real valve: a spring returns it.

The Roller Trip: Exam Symbol

Roller trip

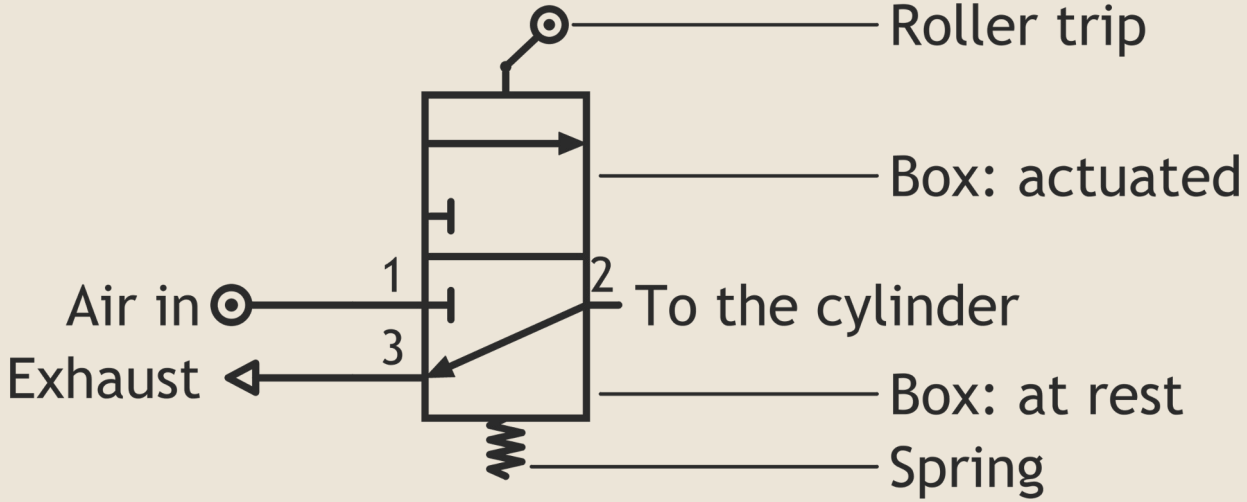
the arm folds one way

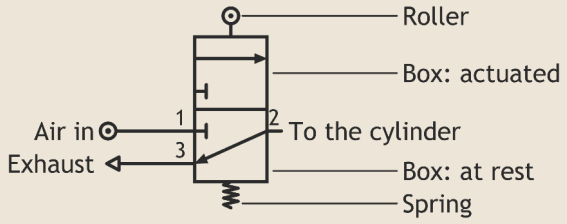
One way

it presses: the valve works

Other way

it folds: nothing happens





Our kit's roller works both ways.

Interlock

Interlock

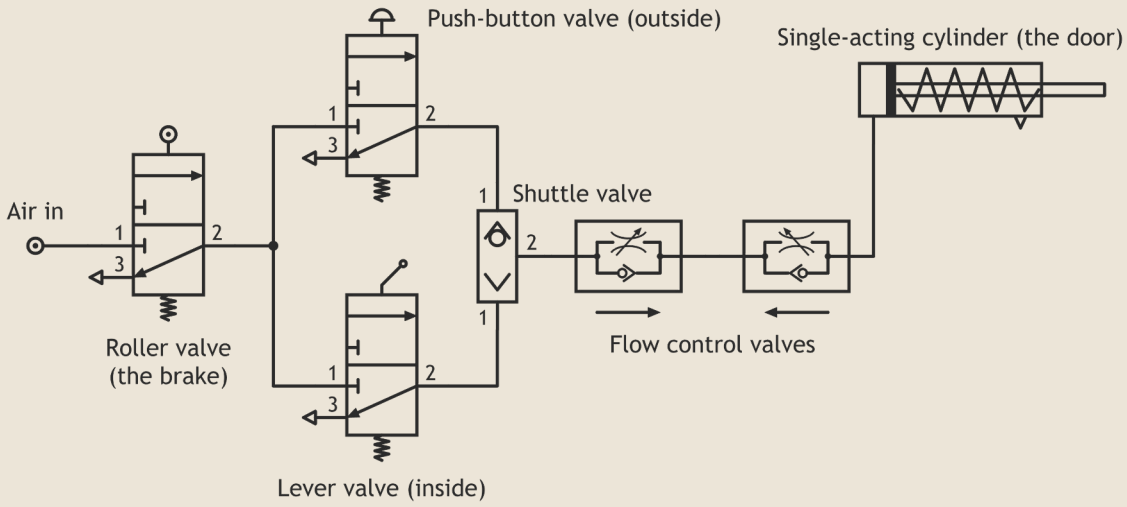
A must be on before B works

Here

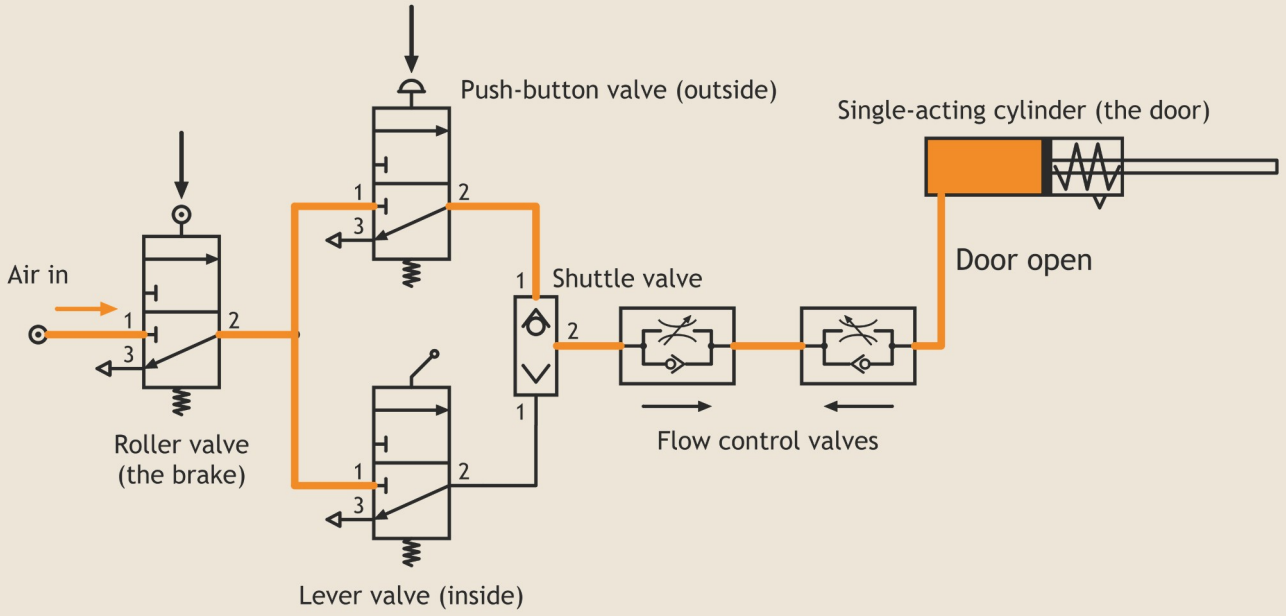
brake on, then the door

Why

it never opens while moving



Brake First: What You Found



1 Brake off, press the button: air stops at the roller.

2 Brake on: air reaches the button and the lever.

3 Brake on, press the button: the door opens.

The Tee Piece

Tee piece

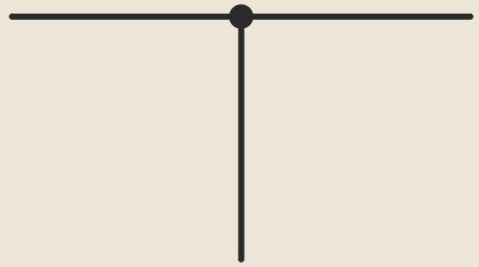
joins three pipes

Symbol

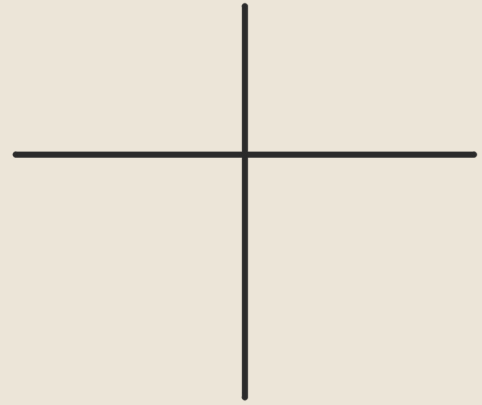
a dot where pipes join

No dot

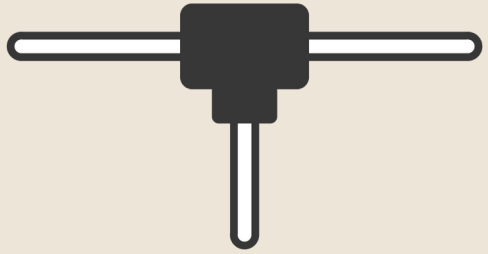
they cross: not joined



Dot: joined (a tee piece)



No dot: they cross

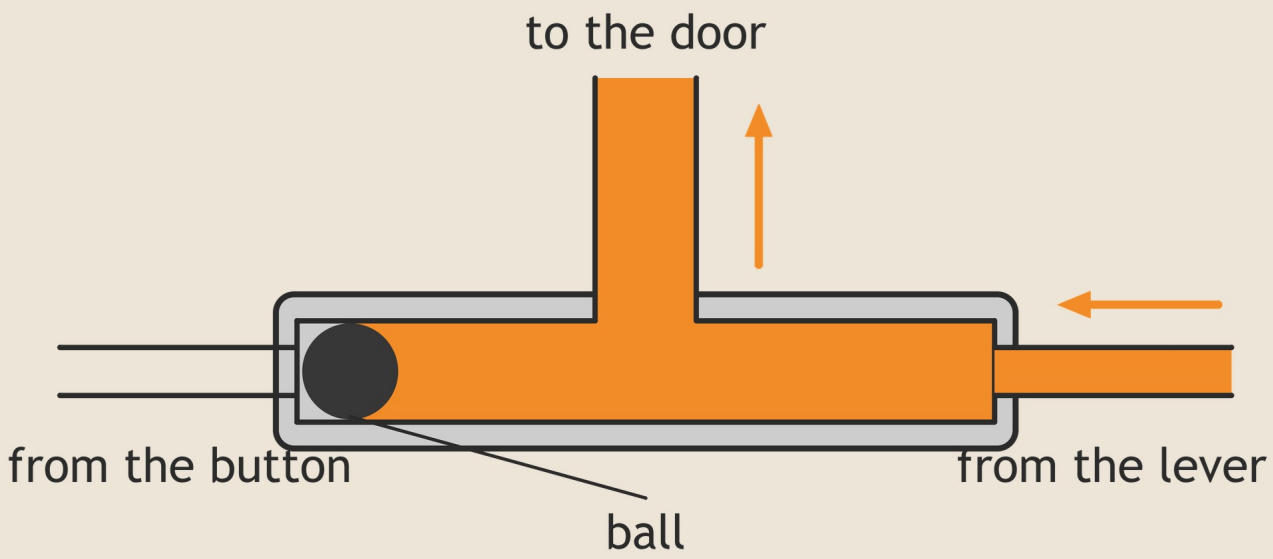


The real T-connector.

Show Me: Name the Symbol

1	A circle on top of the valve box.	Roller
2	A roller arm that folds one way.	Roller trip
3	A dot where three pipes meet.	Joined: a tee piece
4	Two pipes cross with no dot.	Not joined

Inside the Shuttle Valve



1 Air from the button pushes the ball right. It blocks the lever side: air goes to the door.

2 Air from the lever pushes the ball left. Air still goes to the door.

The Shuttle Valve

Inputs

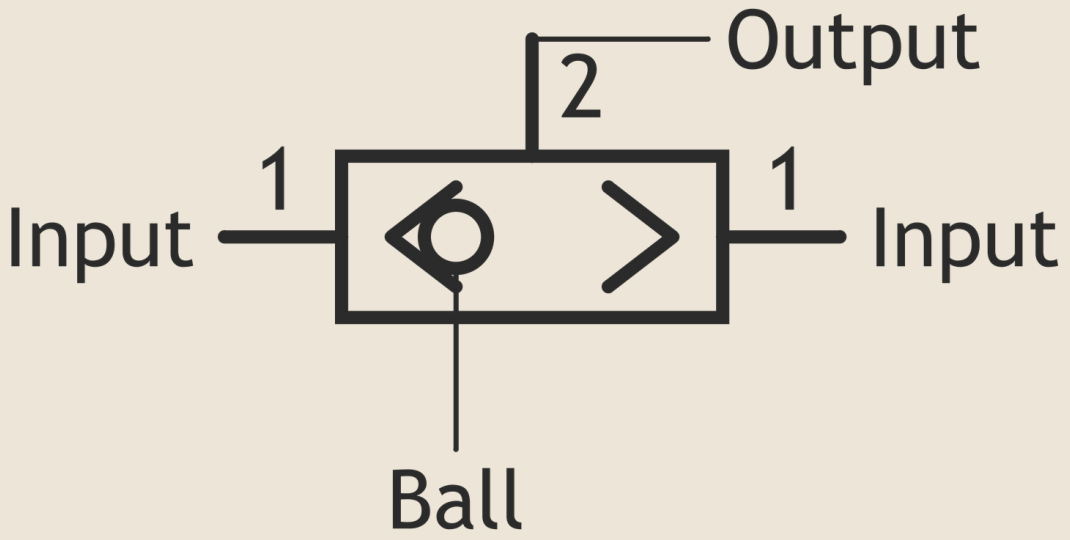
two, both port 1

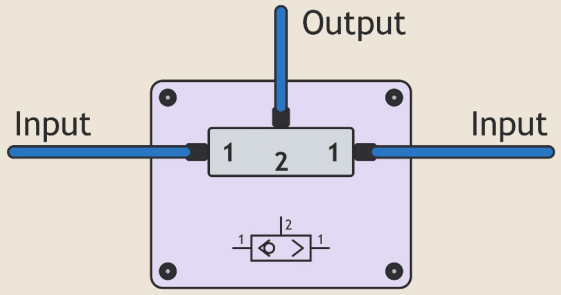
Output

one, port 2

Inside

a ball blocks the side with no air





The real shuttle valve.

OR Control

OR

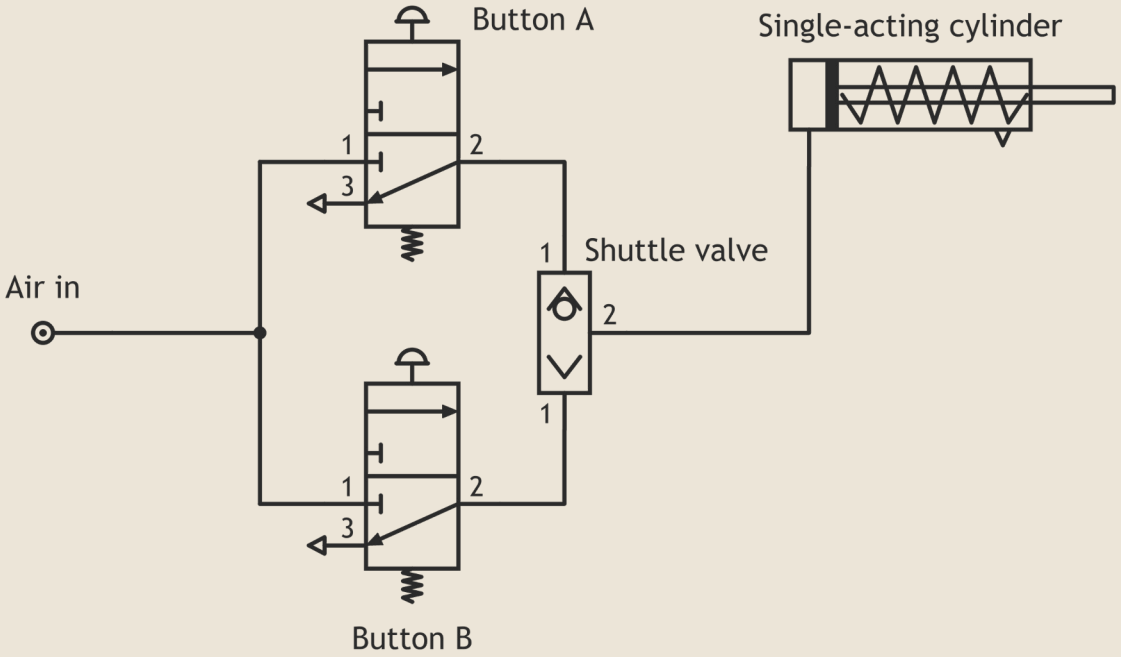
valve A OR valve B works it

Pipes

side by side, into a shuttle valve

Our door

the button OR the lever



AND Control

AND

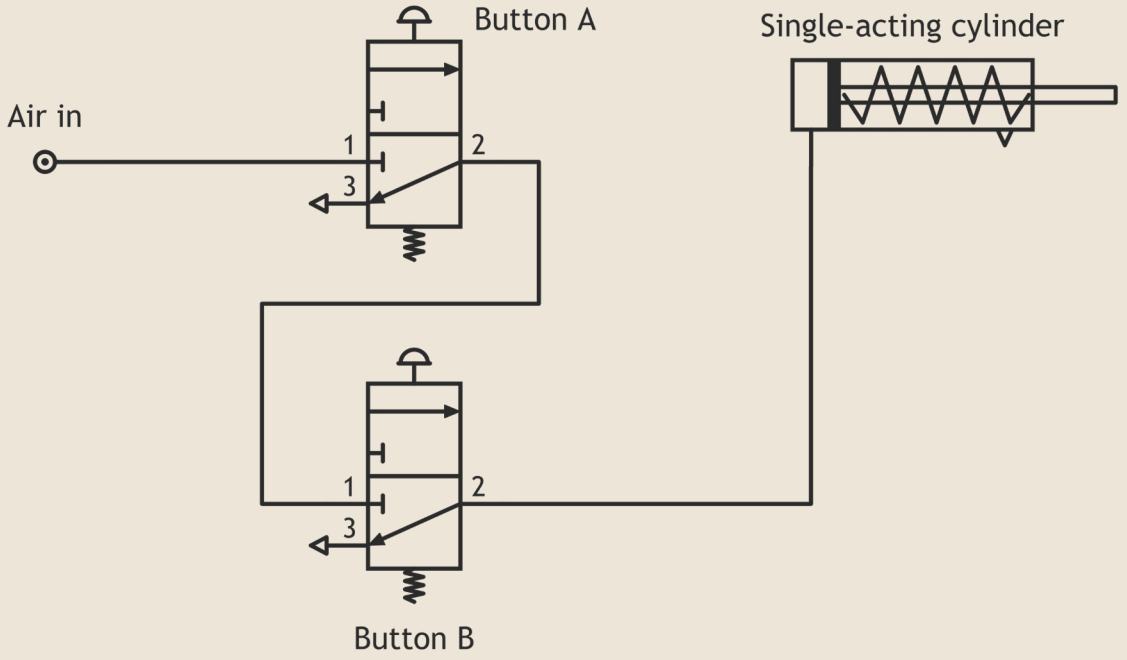
valve A AND valve B, both on

Pipes

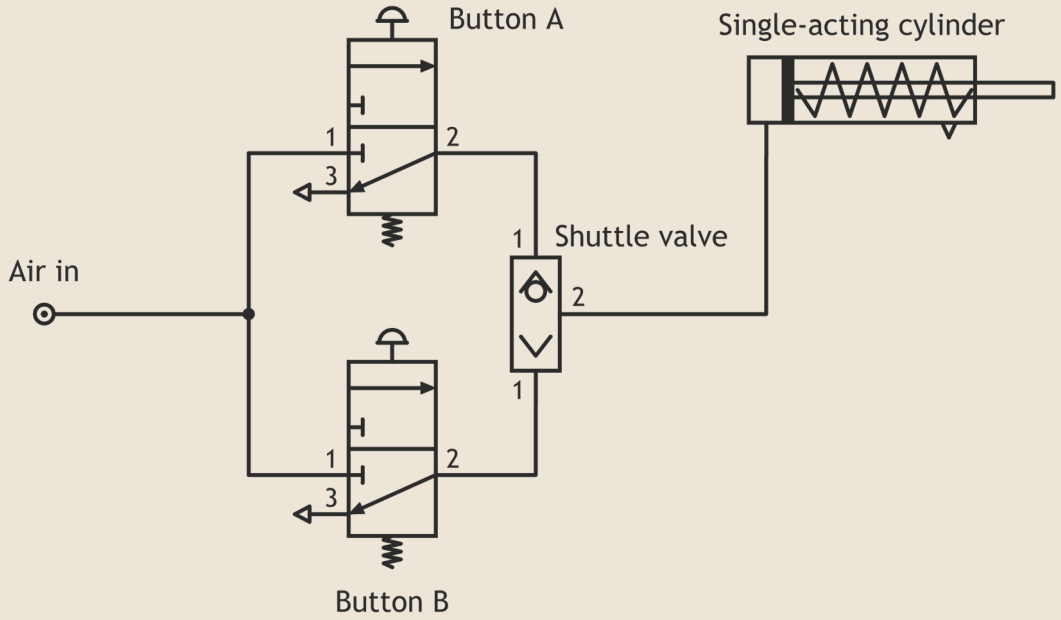
in a line: out of A, into B

Two-hand safety

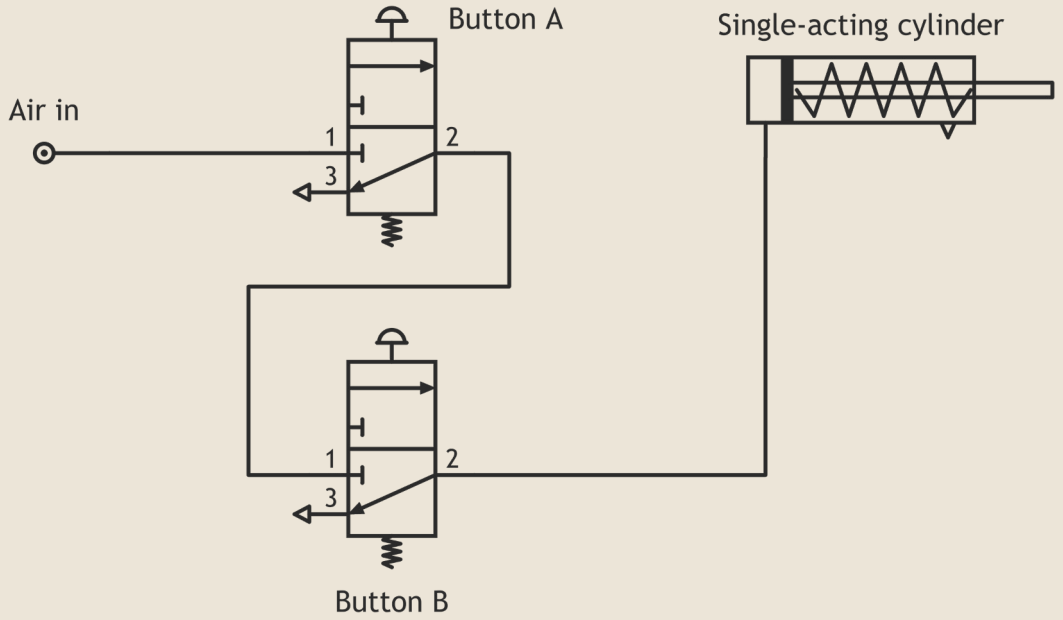
both hands on buttons



Same Valves, Different Pipes



OR: side by side



AND: in a line

We Do: OR and AND



Button A	Button B	OR: rod	AND: rod
-	-	In	In
Pressed	-	Out	In
-	Pressed	Out	In
Pressed	Pressed	Out	Out

Our Door Uses Both

Brake AND a valve

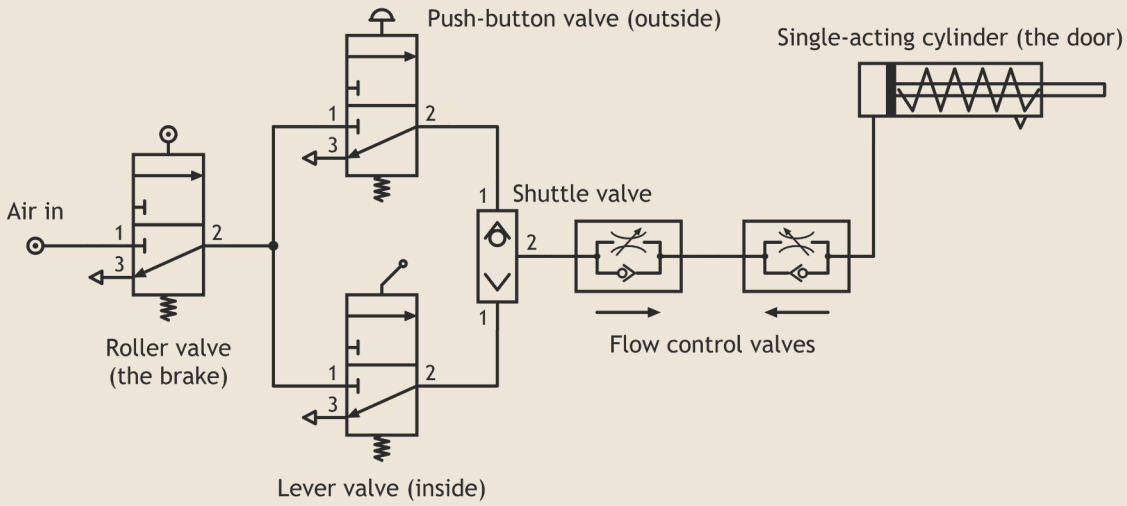
the roller first, in a line

Button OR lever

the shuttle valve

Open when

brake on AND (button OR lever)



Show Me: Which Control?



1	Side by side, joined by a shuttle.	OR
2	In a line: out of one, into the next.	AND
3	A press needs both hands on buttons.	AND
4	A door opens from inside or outside.	OR
5	Opens only if the brake is on too.	AND

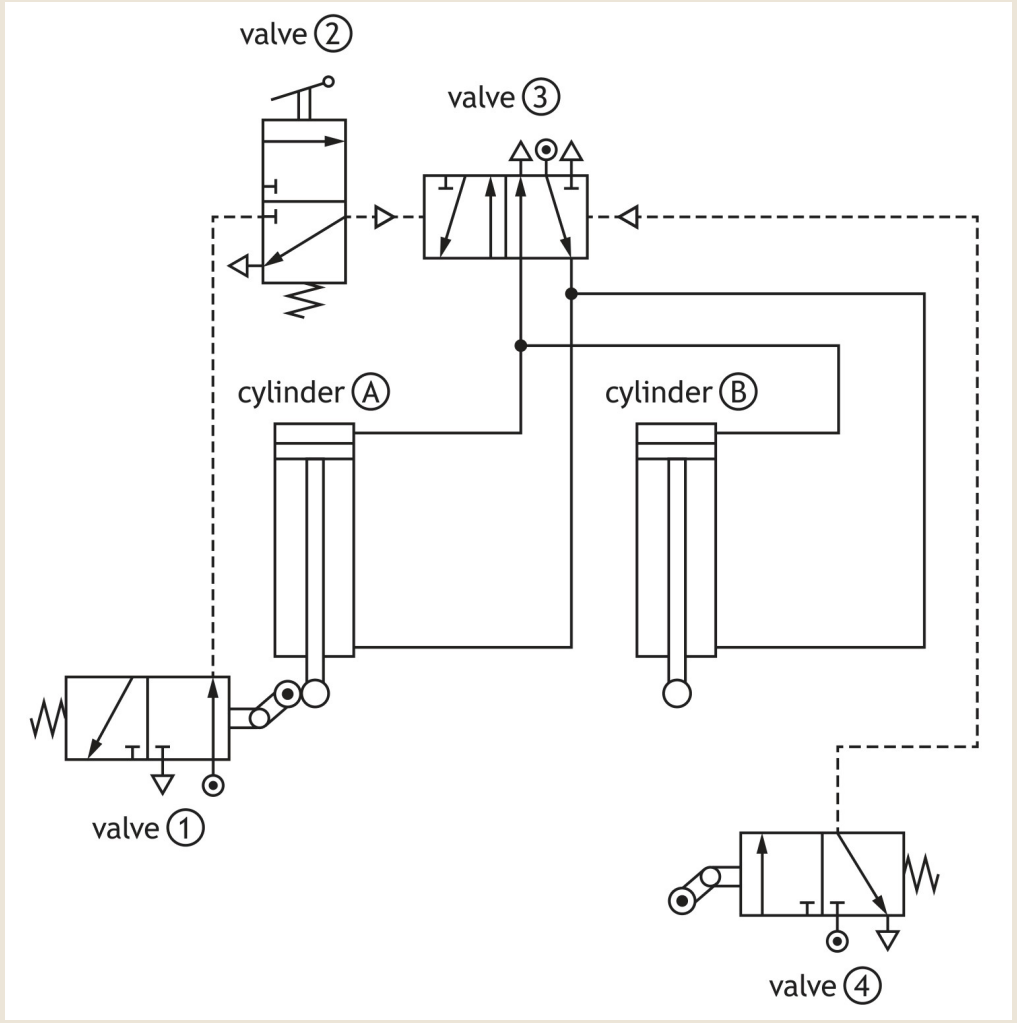
Exam Question: 2025 Q8(a) (1 mark)

Level Up

State the type of control created by the piping of valves 1 and 2.

Hint

follow the dashed pipe out of valve 1



AND (control) (1)

Where You Meet It: Train Doors

Two buttons

inside OR outside

Only works

when the doors are released

Like

our train door



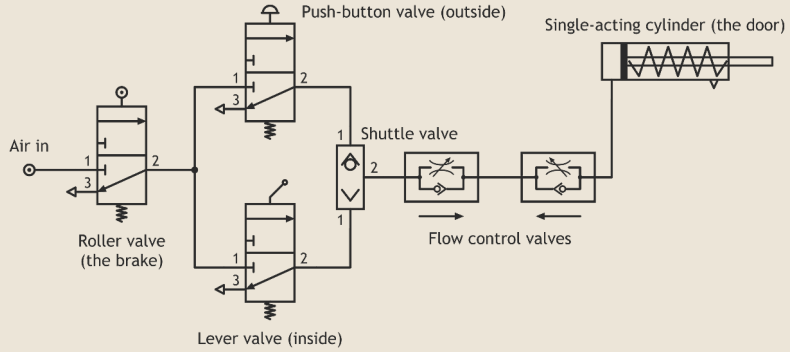
Show Me: Describe the Door (3 marks)

When the roller valve is actuated …

Stage 1 (1): the button is actuated and the piston outstrokes.

Or (1): if the lever is actuated, the piston outstrokes.

Stage 2 (1): when it is released, the valve resets and the piston instrokes.



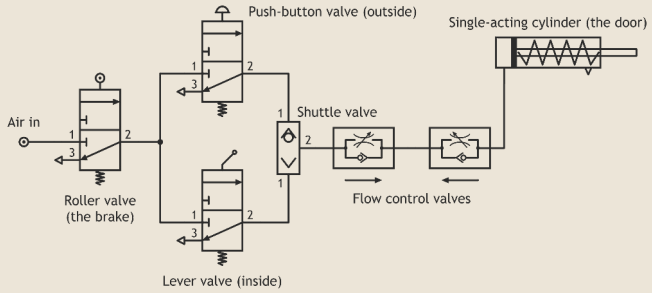
Questions and Key Words

Tasks 6-7 (B6A-L5-p17)

Task 6: Questions 1 to 3 on your own.

Task 7: Key words.

Then page 20, Task 5, Question 3.



7 min

Answers: Tasks 6-7 (B6A-L5-p17)

Question	Mark 1	Mark 2
Q1 (2 marks)	Roller blocks the air	No air to the valves
Q2 (2 marks)	Either input gets out	Ball blocks the other
Q3 (1 mark)	Double-acting cylinder	-
p20 Q3 (1 mark)	Shuttle valve	-
Key words	1 shuttle, 2 OR	3 AND, 4 on

Plenary: How Does the Door Decide?

1	Brake on before the door works. The word?	Interlock
2	Two inputs, one output, a ball inside?	Shuttle valve
3	Two valves into a shuttle: which control?	OR
4	Two valves in a line: which control?	AND
5	Describe: one mark for each…	Stage: valve and piston

Look back — by the end of this lesson…



1. I can explain how the brake stops the door
2. I can tell OR control from AND control
3. I can name the roller, roller trip and tee symbols
4. I can describe how the door works, stage by stage

LOOKING AHEAD

Lesson 9: The Big Gate

Small push, big valve: can air press a button for you?

Credits

Real part drawings: Sources: kit drawings by R Stewart, from the Matrix CP2080 kit.

Train door circuit: The train door circuit adapted from the Matrix CP2080 Pneumatics Essentials Worksheet 3 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Exam Question: 2025 Q8(a): © Qualifications Scotland (SQA). N5 Engineering Science 2025, Question 8(a): the circuit and the marking instructions.

Where You Meet It (train door): Photo: Oxyman, CC BY 2.5, via Wikimedia Commons.

Describe the door: © Qualifications Scotland (SQA). Describe-the-operation layout from the N5 Engineering Science marking instructions (2021 Q12(a), 2025 Q8(b)).

R Stewart - October 2026